



MSE 2202B FINAL PROJECT DESIGN REPORT

Lab 4 Team 5

Team Members

Jason Pan
Liam Mcfadden
Anurag Aggarwal
Alan Sanjit

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Executive Summary

This project involved the design, manufacturing, and testing of an autonomous robotic vehicle capable of collecting, sorting, and returning erasers within a constrained environment. The system was required to operate within a 120 second time limit while meeting constraints related to autonomy, cost, and manufacturability.

An iterative design approach was used, where multiple testing vehicles were developed to identify and resolve design issues. Early challenges in manufacturability, assembly, and subsystem integration were addressed through modular design, standardization of M3 fasteners, and improved 3D printing practices.

The final design, Showcase Vehicle v2, consisted of 17 custom 3D printed components with dimensions of $482 \times 192 \times 255$ mm and a mass of approximately 900 g. The pick up and sorting systems performed reliably, with the sorting system achieving 100 percent accuracy during the showcase, satisfying the primary objective of prioritizing sorting accuracy.

The system also met cost and assembly objectives. The total external cost was approximately \$9, and the use of heat set inserts and standardized fasteners improved serviceability and integration.

The main limitation was the navigation system. A time based control approach was used due to late design constraints, which introduced positional error and prevented full autonomy. As a result, the vehicle required human assistance and did not meet autonomy requirements.

Overall, the project demonstrated effective iterative design. While full autonomy was not achieved, the final system met most objectives and provided a reliable solution for eraser collection and sorting.

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1.1.0 Mission Overview

In fields such as nuclear cleanup, fire fighting, and active mining operations, there are many environments that are too dangerous for humans to operate in. To address this, robotic systems are being developed to enter these areas and retrieve objects of commercial value in their place. This project involves designing and building one of these systems from the ground up.

The vehicle must start at a base location, navigate a flat open area of up to 15 m², find and collect valuable erasers identified by their colour, and return them to a 13 x 10 x 8 cm collection container at the base. The whole task must be completed within 120 seconds. Once started, the robot operates entirely on its own with no human input.

1.2.0 Scoring Analysis

The mission is scored based on an equation involving time (t), number of valuable erasers (g), the number of waste erasers (b), and the weight of the vehicle (w).

$$\text{Score} = \frac{(200 - t) \max((g - 2b), 0)^2}{w}$$

Figure 1: Scoring Equation

The scoring criteria were reviewed early in the design process, and several key implications were identified:

- The score is negatively impacted by vehicle weight. As a result, Objective O-3 was defined to minimize overall robot weight.
- The score is also negatively affected by mission completion time. This led to the inclusion of Objective O-10 to minimize total mission time.
- A penalty is applied for collecting incorrect eraser types. Given the fixed 2-minute time limit, this places greater importance on sorting accuracy over total collection quantity.

1.3.0 Constraints & Objectives

The following outlines the final goals and constraints of the project. Additional goals were introduced throughout the design process based on manufacturing limitations and insights gained from iterative development.

Table 1: Constraints

ID	Constraint	Description
C-1	Size	The robot must be able to fit in the MSE lab locker.
C-2	Sorting of Erasers	The picked-up erasures must be sorted either on the robot or on a stationary sorting system, and must be done based on wavelength (colour).
C-3	Navigation	The robot must be able to traverse the area of operations without any hindrance or help from external sources.
C-4	Power/Battery	The robot must be able to perform its entire task on a single charge of the batteries used, and the batteries used must not be modified.
C-5	Weight Drivable	The robot must be able to move on its own weight and be light enough to be carried by a human (< 3 kg).
C-6	Time-Efficient	Perform the entire operation within the limit of 2 minutes.
C-7	Autonomous	Operations must be performed without any human intervention or wireless communication.
C-8	Budget	The overall cost of components for the robot, excluding components provided by the MSE 2202 lab kit, must not exceed \$50. The overall cost for FDM and laser cutting must not exceed \$20.
C-9	Durability	The robot must be able to perform its task without the expense of its components. The robot must be able to perform the entire operation > 50 times without any need for component replacement.
C-10	Manufacturability	The robot must be manufactured using available materials/components. This includes 4x MSE 2202 lab kit, FDM printed ABS/PETG/PLA components, and external components adhering to the budget constraint.

Table 2: Objectives

ID	Objectives	How to Measure
O-1	Simplicity of Design	The robot should be made with a minimum no. of components to increase reliability and durability.
O-2	Sorting Accuracy & Amount	Sort accuracy (the sorting between high-value and low-value erasers) should be > 85%.
O-3	Efficient Weight Distribution	The robot should be balanced and must not stumble under its own weight and be as light as possible.
O-4	Navigation Accuracy	Should traverse the operation without losing sight of goals and have at least 90% success rate on test runs.
O-5	Efficient UI	Clear indication of each process step and a failsafe in case of incidents or wrong performance of the task
O-6	Budget	The total cost of external components should be < \$20 provided by the project.
O-7	Design for Serviceability	The design shall prioritize ease of maintenance by ensuring that frequently failing or high-wear components can be accessed, removed, and replaced without full system disassembly. Critical electronic components shall remain accessible in the fully assembled configuration, minimizing downtime during troubleshooting or repair. Modular design principles should be implemented, allowing individual subsystems to be serviced independently.
O-8	Design for Assembly	The product shall be designed for straightforward assembly using standard tools and minimal steps. Components, particularly 3D printed parts, shall be designed as modular subassemblies that can be repeatedly assembled and disassembled without degradation. Fastening methods should promote repeatability and alignment, reducing assembly errors and build time.
O-9	Standardization	The design shall utilize M3 screw with M3 5 x 4 mm heated inserts wherever possible to simplify manufacturing, assembly, and maintenance. Use of non-standard fasteners shall be avoided unless functionally justified (e.g., load requirements, space constraints). General dimensional tolerances shall be maintained at ± 0.5 mm to ensure compatibility between interfacing components while accommodating typical manufacturing variability.
O-10	Speed	The amount of time per mission should be minimized < 1 minute.

2.1.1 Design Decision Structure

For the conceptual design phase, an elimination based approach was used to determine subsystem configurations. The overall system was decomposed into 6 major subsystems, each with a defined function:

- Drive System: Responsible for all vehicle motion, including propulsion and turning.
- Chassis: Provides structural support and integrates all subsystems into a single assembly.
- Drop System: Responsible for delivering collected erasers into the scoring bin.
- Sorting System: Determines and directs sorts erasers into the correct bin.
- Pick-Up Sensing / Navigation System: Responsible for vehicle navigation and detection of erasers.
- Pick-Up Mechanism: Responsible for the physical collection of erasers.

This decomposition allowed each subsystem to be evaluated and developed independently during concept generation and selection.

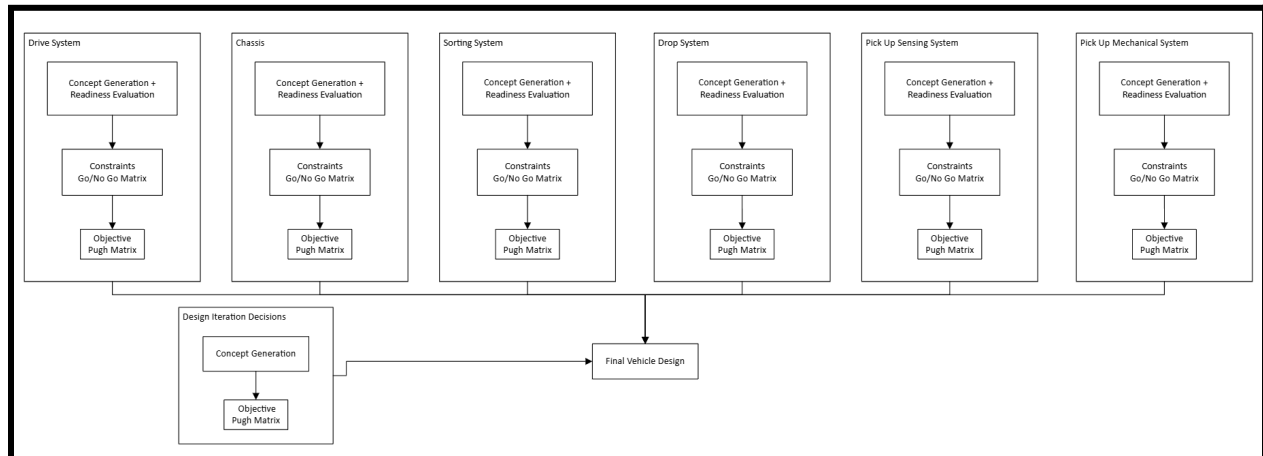


Figure 2: Design Decision Structure Flow Chart

Each subsystem concept first undergoes a readiness evaluation to identify obvious constraint violations or design flaws, allowing weak concepts to be eliminated early. Concepts that pass this stage are then evaluated using a Go/No-Go matrix to ensure compliance with all design constraints.

Following this, remaining concepts are further refined and compared using a Pugh matrix to support final selection of subsystem configurations. Evaluation criteria are assigned weights based on their relative importance, which are then normalized such that their total sums to 1.0. This ensures each criterion contributes proportionally to the final score and prevents any single category from dominating the decision unless intentionally weighted. This improves consistency and allows for clearer justification when comparing concepts across different metrics.

For design issues identified after the conceptual phase (e.g., during preliminary design or manufacturing), an expedited evaluation process is used. In these cases, concepts are assessed using a simplified Pugh matrix without weight normalization and parameters will be graded from -2 to 2. This reduces evaluation time and allows for faster implementation within the limited time between milestone deadlines, at the expense of reduced analytical rigor.

2.1.2 Trade Studies

Table 3: Concept Generation & Readiness Evaluation

ID	Concept	(Keep/Drop)	Reasoning
Robot Movement			
1.	Gears with Tracks	Keep	
2.	Aftermarket Rubber Wheels	Keep	
3.	Self-Manufactured 3D Printed Wheels + Rubber Band Tires	Keep	
Chassis Design			
1.	4 Wheel Drive	Keep	
2.	2 Wheel Drive	Keep	
	S Wheel Drive + Dedicated Steering	Drop	Too complex with limited benefits.
Eraser Sorting			
1.	Conveyor Belt + 2 Lane Knocking System	Keep	

2.	Steep Slide with Hatch System	Keep	
3.	Slide With Sorting Wheel	keep	
4.	Angle Slide with Sensor Knocking System	Drop	It requires very precise timing and multisystem cooperation.
Eraser Drop Mechanism			
1.	Slide	Keep	
2.	Conveyor Belt	Drop	Require too much power and components.
3.	Gate	Keep	
Eraser Pick Up Sensing			
1.	Camera	Drop	ESP32 does not have enough computational resources for image processing.
2.	UltraSonic Sensor	Keep	
3.	Infrared Sensor	Keep	
4.	UltraSonic + Infrared Sensor	Keep	
Eraser Pick Up Mechanism			
1.	Scoop pick-up	keep	
2.	Sticky Conveyor Belt	Drop	Stickiness might lose its effectiveness over time.
3.	Claw	keep	
4.	Vacuum Suction	Drop	Require many additional components on the robot.
5.	Stabbing needle arm	Drop	Requires too much precision, too slow.
6.	Scoop pick-up + Gate	Keep	

Table 4: Go/No-Go Matrix

ID	Concepts	Size	Sorting of Erasers	Navigation	Power/Battery	Weight Drivable	Time Efficient	Autonomous	Budget	Durability	Manufacturability	Reasoning
1	Gears with Tracks	Go	N/A	Go	N/A	Go	N/A	N/A	Go	Go	Maybe	- Requires precise alignment, tension control, and additional mechanical components, increasing fabrication complexity. - Higher material cost reduces overall efficiency and mobility compared to simpler wheel-based systems.
2	Aftermarket Rubber Wheels	Go	N/A	Go	N/A	Go	Go	N/A	Maybe	Go	Go	- Purchasing pre-manufactured components increases total project cost. - Cost may not justify marginal traction improvement over lower-cost alternatives.
3	Self-Manufactured 3D Printed Wheels + Rubber Band Ties	Go	N/A	Go	N/A	Go	N/A	N/A	Go	Maybe	Go	- 3D printed materials may wear down under continuous friction and load. - Rubber bands can stretch or degrade, reducing long-term reliability.
4	4 Wheel Drive	Go	N/A	Go	No Go	Go	Go	No Go	Go	Go	Go	- Requires synchronization between two ESP32 boards, increasing software complexity. - Coordinating encoder feedback and motor control adds unnecessary programming overhead.
5	2 Wheel Drive	Go	N/A	Go	Go	Maybe	Maybe	Go	Go	Go	Go	- Uneven weight distribution may reduce traction and stability. - Limited drive contact points can affect load handling performance.
6	Conveyor Belt + 2 Lane Knocking System	No Go	Go	Go	Go	Go	Maybe	Go	Maybe	Maybe	Maybe	- System footprint exceeds locker size constraints. - Additional belt and lane structure increases spatial requirements.
7	Steep Slide with Hatch System	Go	Go	N/A	Go	Go	Go	Go	Go	Go	Go	
8	Slide With Sorting Wheel	Go	Go	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Sorting wheel timing must align precisely with item arrival. - Rotational misalignment may reduce throughput efficiency.
9	Slide	Go	N/A	N/A	Go	Go	Go	Go	Go	Go	Go	
10	Gate	Go	N/A	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Servo actuation introduces delay during opening and closing cycles. - Mechanical lag may slow repeated sorting operations.
11	Ultrasonic Sensor	Go	Go	Go	Go	Go	Go	Go	Go	Go	N/A	
12	Infrared Sensor	Go	Go	No Go	Go	Go	Go	Go	Maybe	Go	N/A	- Lab IR sensor is unreliable for accurate distance measurement. - Readings are affected by ambient light and surface reflectivity.
13	Ultrasonic + Infrared Sensor	Go	Go	Go	Go	Go	Go	Go	Go	Go	N/A	
14	Scoop pick-up	Go	Go	N/A	Go	Go	Go	Go	Go	Go	Go	
15	Claw	Go	Go	N/A	Go	Maybe	Maybe	Go	Go	Maybe	Maybe	- Requires precise servo calibration and grip control. - Complex geometry increases fabrication difficulty.
16	Scoop pick-up + Gate	Go	Go	N/A	Go	Go	Maybe	Go	Go	Go	Go	- Combining scoop and gate may introduce redundant handling steps. - Additional mechanism coordination increases cycle time.

Table 5: Pugh Matrix Weight Normalization

Objectives	Simplicity of Design	Sorting Accuracy & Amount	Efficient Weight Distribution	Navigation Accuracy	Efficient UI	Budget	Total	Normalized Weighting	Legend
Simplicity of Design	1.00	0.14	1.00	0.20	9.00	0.14	11.49	0.12	Relative Importance
Sorting Accuracy & Amount	7.00	1.00	7.00	1.00	9.00	1.00	26.00	0.27	Exceptionally Less
Efficient Weight Distribution	1.00	0.14	1.00	0.20	9.00	0.33	11.68	0.12	Much Less
Navigation Accuracy	5.00	1.00	5.00	1.00	9.00	0.20	21.20	0.22	Less
Efficient UI	1/9	1/9	1/9	1/9	1.00	0.11	1.56	0.02	Slightly Less
Budget	7.00	1.00	3.00	5.00	9.00	1.00	26.00	0.27	Equally Important
							97.92	1.00	Slightly More
									More
									Much More
									Exceptionally More

Table 6: Pugh Matrix

#	Objective	How to Measure	Weight (higher number is more important)	Gears with Tracks	Aftermarket Rubber Wheels	Self-Manufactured 3D Printed Wheels + Rubber Band Ties	2 Wheel Drive	Steep Slide with Hatch System	Slide With Sorting Wheel	Slide	Gate	Ultrasonic Sensor	Ultrasonic + Infrared Sensor	Scoop pick-up	Claw
1	Simplicity of Design		0.12	-1	2	1	1	1	2	2	-1	0	-2	2	-1
2	Sorting Accuracy & Amount		0.27	0	0	0	2	2	2	0	0	0	0	0	0
3	Efficient Weight Distribution		0.12	2	1	1	0	-1	0	0	-1	0	0	-1	-1
4	Navigation Accuracy		0.22	-1	1	1	2	0	0	0	0	0	1	1	-1
5	Efficient UI		0.02	0	0	0	0	0	0	0	0	0	2	2	0
6	Budget		0.27	-1	-1	2	2	1	2	2	1	2	2	2	-1
Results															
Weighted Totals				-0.360848139	0.304824277	0.984113609	1.03416779	0.794644015	1.292719971	0.765059447	0.0289845675	0.5310595254	0.3447412787	0.4788069529	-0.7185838413

Table 7: Material Trade Study

Metric	Weight	Laser Cut Cardboard	3D Printed	Handcrafted Cardboard
Manufacturing Complexity	25.00%	2	1	-2
Weight	15.00%	2	-1	1
Design Complexity	30.00%	-2	0	1
Reliability	20.00%	-1	2	-2
Cost	10.00%	1	-1	2
Total	100.00%	0.1	0.4	-0.25

Material selection was performed after the conceptual phase as a design iteration decision using a simplified trade study. 3D printing was selected as it scored the highest and offered high accessibility, with 2 team members having direct access to printers, enabling faster iteration and prototyping.

Table 8: Navigation System Trade Study

Metric	Weight	IR Becon	Ultrasonic Sensor + Wall	IR Beacon + Ultrasonic Sensor + Wall
Reliability	40.00%	-1	1	2
Software Complexity	50.00%	-1	2	-2
Integration Complexity	10.00%	1	0	-1
Total	100.00%	-0.8	1.4	-0.3

The navigation system metrics were revised after the conceptual phase due to an incorrect assumption about the capabilities of the infrared sensor. This was treated as a design iteration decision. Initially, the infrared sensor was assumed to be able to detect eraser colours and assist with navigation. Testing showed this was not feasible, as the sensor could not reliably differentiate colours in this application. As a result, the navigation trade study was redone based on realistic sensor performance. The ultrasonic sensor with a surrounding wall was ultimately selected, as it provided more reliable detection and navigation.

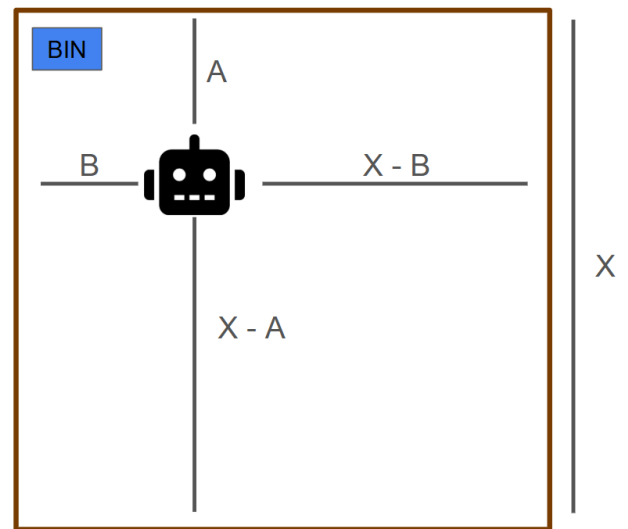


Figure 3: Ultrasonic Sensor + Wall Navigation System

1.4.0 Project Timeline

A project timeline was established following Milestone 1 to guide the development of both the testing and showcase vehicles. The work was divided into three phases, each consisting of tasks scheduled based on expected completion time and milestone deadlines. Subsystems were developed in parallel where possible, while still accounting for dependencies such as integration and manufacturing. The Gantt chart reflects this through overlapping design, manufacturing, and testing, with time allocated for iteration.

1. “Preliminary Design Phase”: This phase focuses on developing the core subsystems. Work includes material trade studies and initial CAD for the chassis, drive system, navigation system, and electrical layout. Most CAD is completed here and refined to a manufacturable level. Subsystems are developed in parallel while considering integration. By Milestone 2, major design decisions are finalized and the design is ready for manufacturing.

2. “Detailed Design Phase”: This phase focuses on physical implementation and refinement. The testing vehicle is manufactured first, followed by development of a showcase vehicle which includes subsystems such as pick-up, sorting, and chassis integration. Issues such as fitment, alignment, and tolerance stack-up are expected, and time is allocated for iteration. Manufacturing also establishes standards for tolerances, fasteners, and assembly. Software is developed alongside hardware integration. This phase ends with a fully assembled system (Milestone 4).

3. “Testing, Refinement, and Evaluation Phase”: This phase focuses on validating performance and making final adjustments. Subsystems are tested individually before full system testing. Results are compared against design expectations, and issues are addressed through targeted refinements. The phase ends with full system integration and a final showcase under expected operating conditions, ensuring the system is reliable and consistent.

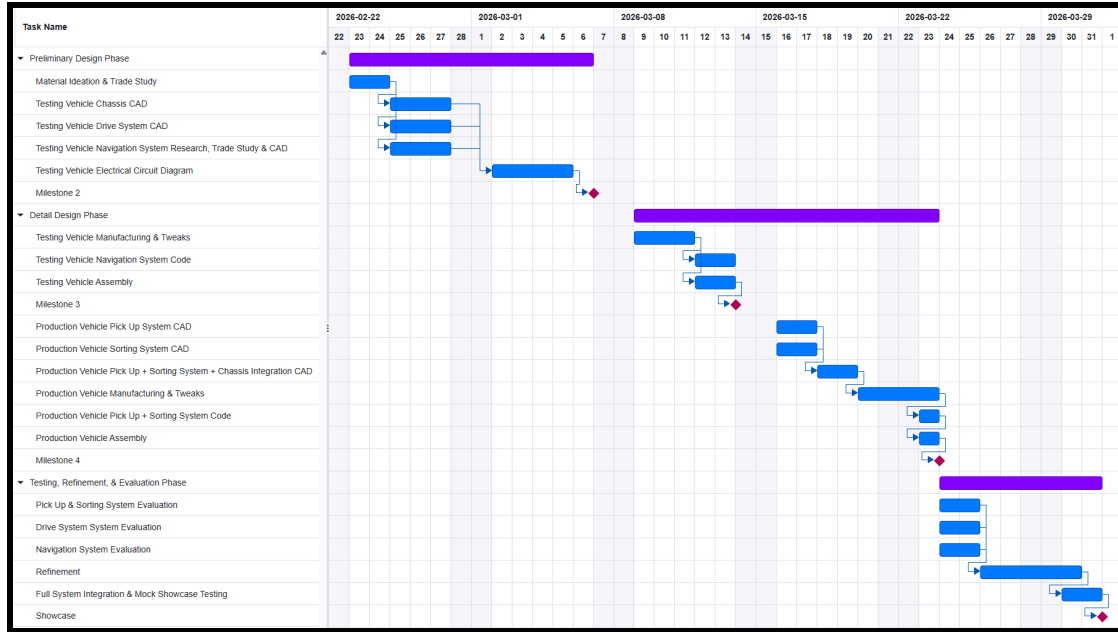


Figure 4: Gantt Chart for Project Timeline

2.2.1 Testing Vehicle Manufacturing

To begin the first attempt of the testing vehicle (Designation: Testing Vehicle v1), the milestone 2 testing vehicle was manufactured using a P1S 3D printer using PETG filament.

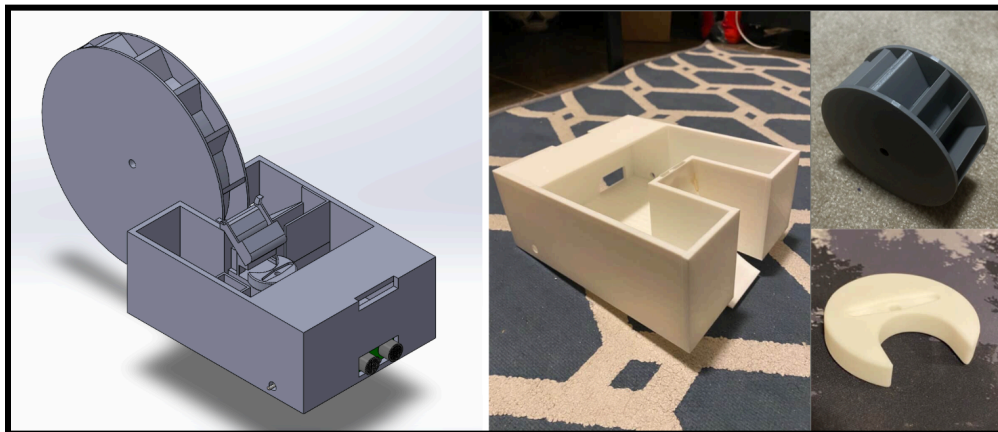


Figure 5: Testing Vehicle v1 CAD & 3D Printed Parts

Following the manufacturing of the initial testing vehicle, several design issues were identified:

1. The design required extensive 3D printing support. Due to multiple overhangs, there was no viable print orientation that significantly reduced support usage, leading to increased print time and post processing.
2. The vehicle was difficult to assemble and service. Many components were designed as single large parts, making internal features hard to access. This also increased iteration time, as small errors required reprinting entire parts.
3. Material usage was inefficient. Excess filament was used due to unnecessarily thick walls and fully solid geometries where weight reduction features (e.g., cutouts) could have been implemented.
4. Part integration was inconsistent. A lack of standardization in connection methods led to poor fit between components and required additional adjustments during assembly.
5. Collaboration in CAD was very limited. Subsystems were developed independently by each team member on local SolidWorks files, making it difficult to track changes across components. This resulted in integration issues and frequent redesigns.
6. Vehicle weight became a concern, particularly regarding the capability of the N20 motors to drive the system. Analytical estimation was attempted but was inconclusive, indicating the need for physical testing.

These shortcomings of the initial design was evaluated and goals O-7, O-8, and O-9 was established along with the following design principles:

- Minimization of 3D Printing Supports
 - All 3D printed components shall be designed to minimize or eliminate the need for support structures, reducing material usage, print time, and post processing effort.
 - Geometries should incorporate self supporting angles ($\geq 30^\circ$ where possible) and avoid overhangs that require excessive support.
- Print Orientation and Structural Integrity

- Parts shall be designed to allow optimal print orientation, ensuring that layer adhesion does not compromise structural performance.
- Load-bearing components must be oriented such that principal stresses are aligned with the strongest print direction (layer plane).
- Design geometry should account for anisotropic material properties inherent to FDM printing.
- Dimensional Tolerancing
 - General dimensional tolerances shall be maintained at ± 0.5 mm to ensure compatibility between interfacing components while accommodating typical manufacturing variability.

2.2.2 Testing Vehicle Testing

To evaluate the capability of the N20 motors, a simplified redesign (Designation: Testing Vehicle v2) was created using Onshape. The design incorporated several updated goals and constraints, and was then printed and assembled. The vehicle was configured with an approximation of the final electrical system by connecting the N20 drive motors directly to a DC power supply, allowing input voltage to be monitored during testing.

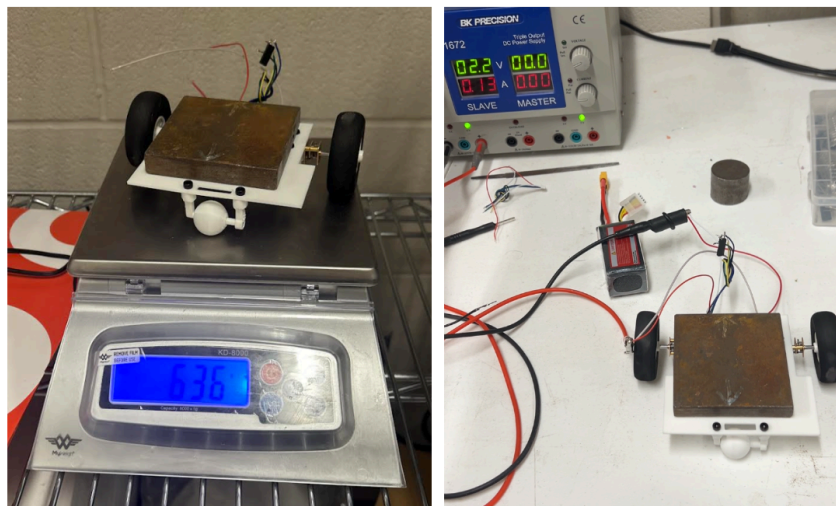


Figure 6: Testing Vehicle v2 Weight & Voltage Testing

A load of ~630 g, representing the predicted final vehicle weight, was applied. Testing showed that the vehicle was able to initiate movement at ~2.0 V, which is below the expected operating voltage of 5 V. Based on this, the N20 motors were determined to be sufficient for the application.

2.2.3 Testing Vehicle Iteration

Building on the foundation established by Testing Vehicle v2, Testing Vehicle v3 was developed for Milestone 3, incorporating the updated goals, constraints, and integration of the navigation system. The design was made significantly more modular, with components separated wherever possible to improve manufacturability and reduce print complexity, especially after the introduction of structural cutouts.

Connections between components were standardized using M3 screws paired with heatset threaded inserts. Heatset inserts were selected due to their ease of installation, as they can be directly embedded into printed parts, and their ability to provide durable, repeatable threads in plastic. Compared to directly threading into printed material, inserts significantly reduce wear and risk of stripping, especially under repeated assembly and disassembly.

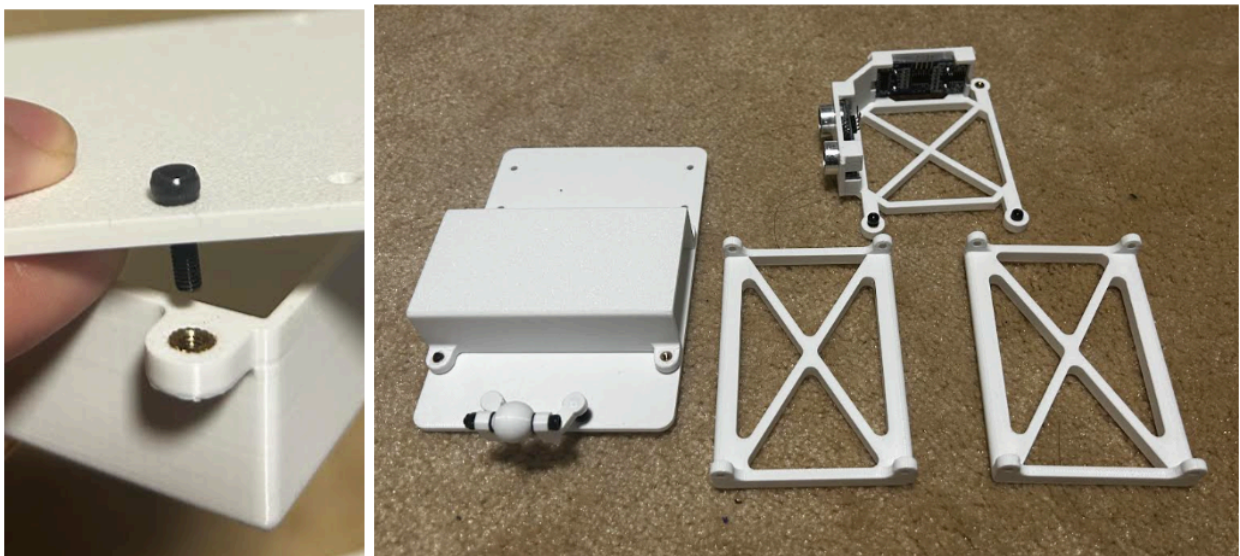


Figure 7: Heated Inserts & Showcase of Design Modularity

This approach improves serviceability and assembly efficiency (Objectives O-7 and O-8), as components can be quickly removed and reattached without degrading the connection. It also eliminates the need for loose nuts, reducing the risk of part loss and simplifying handling in constrained spaces. Additionally, standardizing fasteners across the design improves team collaboration (Objective O-9), as all members work with a consistent interface. This reduces integration issues and allows subsystems to be developed more independently without compatibility concerns.

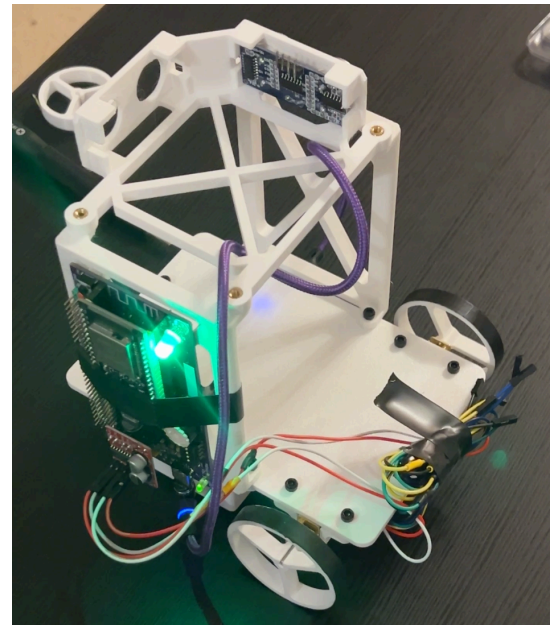


Figure 8: Test Vehicle v2

2.2.4 Showcase Vehicle Manufacturing

Showcase Vehicle v1 initially reused the pick-up system from Testing Vehicle v1. However, during integration and testing, a key design oversight was identified. The lift geometry was too small, such that when the mechanism rotated to the height of the slide, the eraser did not have sufficient clearance or travel to transfer off the lift. As a result, the eraser remained trapped on the lift rather than being guided onto the next stage of the system.

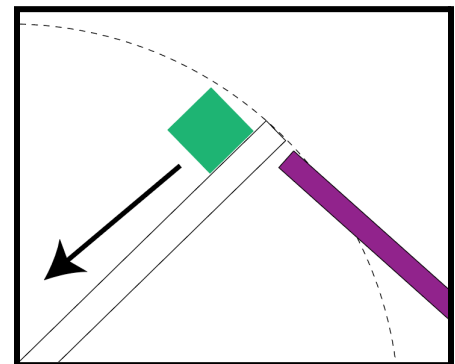


Figure 9: Depiction of Testing Vehicle v1 Lift System Flaw

Two solutions were considered:

- Replace the lift mechanism with a stick-based system, where erasers would adhere to the surface of the lift and be carried from the top side of the rotating wheel

- Increase the diameter of the wheel to approximately 25 cm, allowing sufficient space within the lift geometry for the eraser to naturally slide onto the transfer surface

The stick-based system was ultimately selected. While increasing the wheel size would have resolved the geometric constraint, it would have significantly increased the overall size and mass of the system. The stick system provided a more compact solution and contributed to weight reduction, aligning with Objective O-3. The rotation of the pick-up wheel was implemented using a spur gear system designed through a custom plugin in Onshape. A gear ratio of 1:2 was selected to increase output torque.

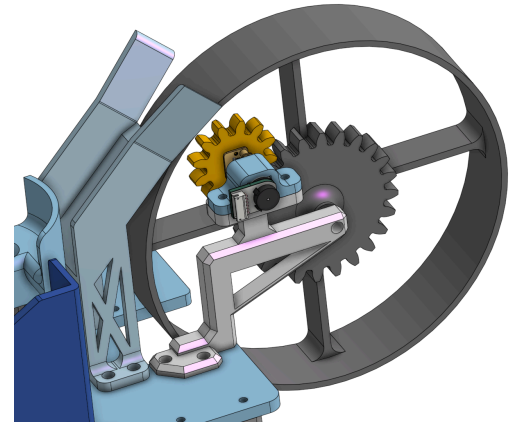


Figure 10: Gear Mechanism of Showcase Vehicle v2

This decision was made instead of directly coupling the motor to the wheel, as the system was not expected to operate under ideal conditions. Resistance from eraser contact, friction, and potential misalignment during pickup were all anticipated, and the increased torque provided by the gear reduction improved reliability under these conditions.

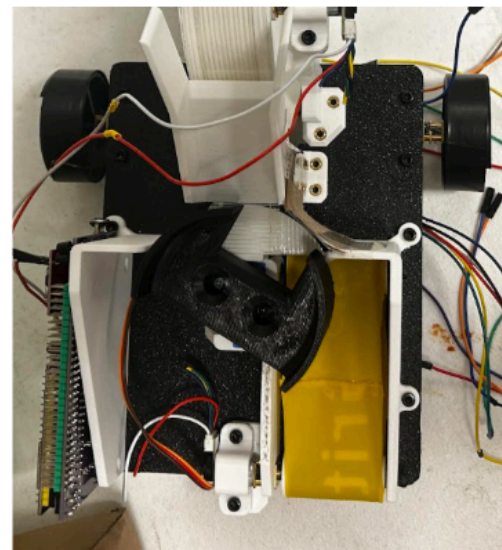
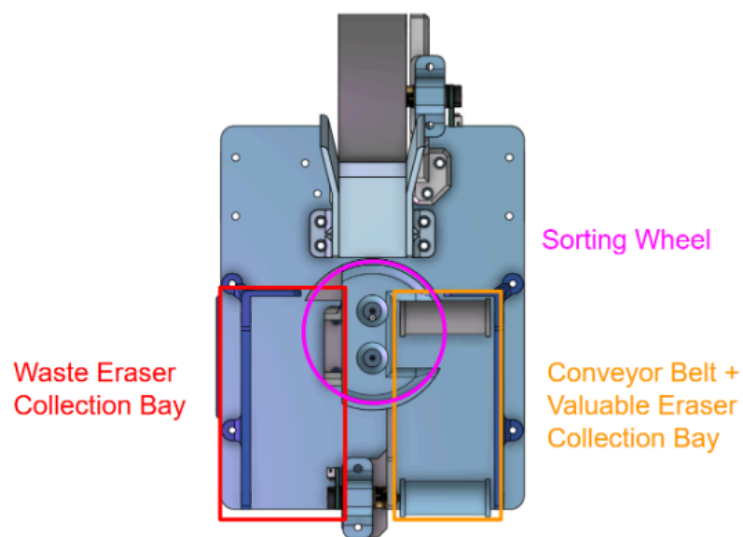


Figure 11: Sorting System & Drop System Design vs. Showcase Vehicle v1

Following this, the pick-up system was integrated with the sorting system. The sorting system consisted of a rotating sorting wheel positioned above two collection bays, one on each side. The intent was to direct erasers into the correct bin based on sorting logic. An eraser drop mechanism using a servo was initially implemented to push erasers from the platform into the bins. However, testing revealed that the servo did not produce sufficient torque to reliably move the erasers, especially under conditions where multiple erasers or increased friction were present. As a result, this approach was deemed unreliable. To resolve this, a conveyor based transport system was designed and implemented. This system moved erasers from the pick-up region, beneath the rotating wheel, and into the appropriate collection bins.

2.2.5 Showcase Vehicle Testing

During testing of Showcase Vehicle v1, multiple issues were identified:

- Problem 1: The eraser slot is too shallow, causing the eraser to become trapped between the walls of the sorter wheel and the slide during rotation. This results in the sorter wheel stalling.



Figure 12: Demonstration of Eraser Getting Stuck 1

- Problem 2: Erasers can become lodged in the gap beneath the sorter wheel, again preventing proper rotation and causing the system to stall.



Figure 13: Demonstration of Eraser Getting Stuck 2

- Problem 3: At the sorted position, the ramp geometry is too large, which in some cases prevents the eraser from falling into the collection area, reducing reliability of the drop system.

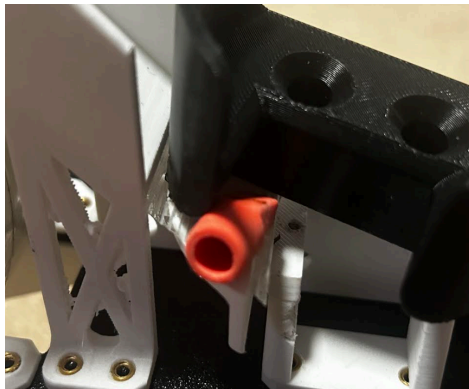


Figure 14: Demonstration of Eraser Getting Stuck 3

- Problem 4: The stick based lift system did not perform as expected. The surface coating of the erasers prevented reliable adhesion to the wheel. Multiple adhesives were tested, including tensile tape, alien tape, and duct tape; however, changes in adhesive type had minimal impact on performance. It was determined that the eraser surface properties were the limiting factor, making this approach inherently unreliable.

2.2.6 Showcase Vehicle Iteration

Based on the testing results, the design reverted to the second option of using the original lift concept with an increased wheel diameter marking the beginning of Show Case Vehicle v2. This change ensured sufficient clearance for the eraser to transfer properly onto the slide. Testing of the sorting system also revealed that the erasers, due to their “gummy” material properties, were highly prone to getting stuck in

small gaps, even at tolerances as low as 0.5 mm. To address this, grooves were introduced into the slide of the pick-up system.

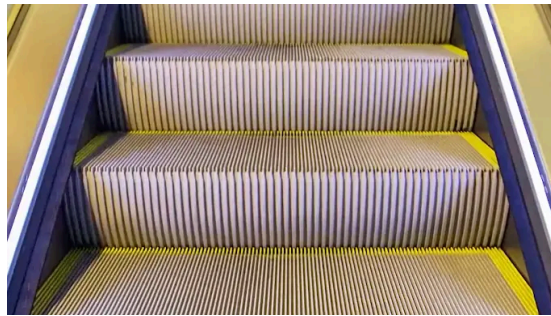


Figure 15: Example of Escalator Groves

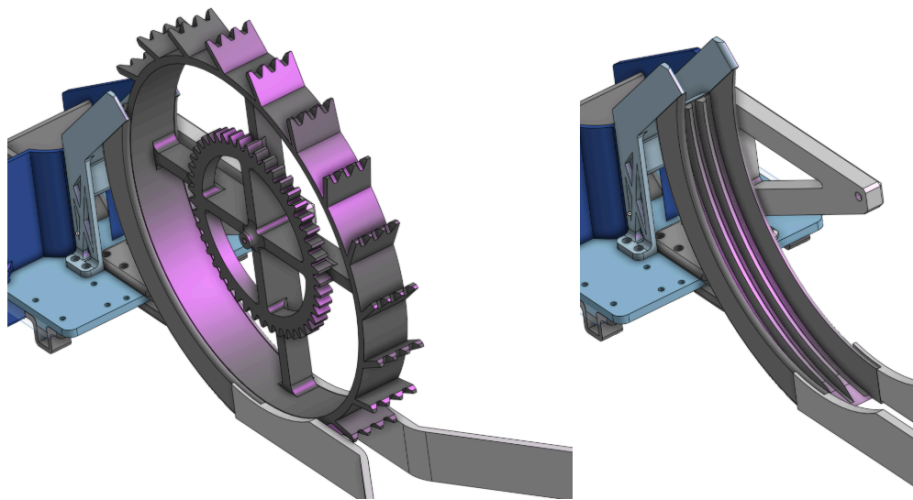


Figure 16: Implementation of Escalator Groves on Showcase Vehicle v2

This approach was inspired by escalator designs, where grooves are used to prevent soft materials, such as shoe soles, from getting caught in gaps. Additionally, the increased wheel diameter resulted in higher torque requirements when overcoming resistance from the erasers. To compensate for this, the gear ratio was increased to 1:4, providing additional torque and improving the reliability of the pick-up mechanism under load.

One issue introduced by the addition of grooves was related to manufacturability. The teeth geometry on the wheel would typically require significant 3D printing support, increasing material usage and post-processing time. Removing support from small features such as teeth is also time consuming and can

negatively affect surface quality. It was also observed from previous iterations that surfaces printed without support consistently resulted in better dimensional accuracy and smoother finishes, while supported regions showed reduced tolerance and rougher surfaces. To address this, the groove and tooth geometry were redesigned to be self supporting in the selected print orientation. The features were angled at approximately 29° relative to the horizontal, keeping them below the typical $\sim 30^\circ$ threshold where support is required. This allowed the benefits of the groove design to be retained without introducing additional support material or compromising print quality.

For the sorting system, a slope was implemented on the landing platform so that the erasers will fall more easily to the collection bay when pushed by the sorter wheel, mitigating problem 3. The height of the slide was also raised so that it matched the height of the eraser; mitigating problem 1. Last, the landing platform was raised to decrease the space between it and the sorting wheel; mitigating problem 2.

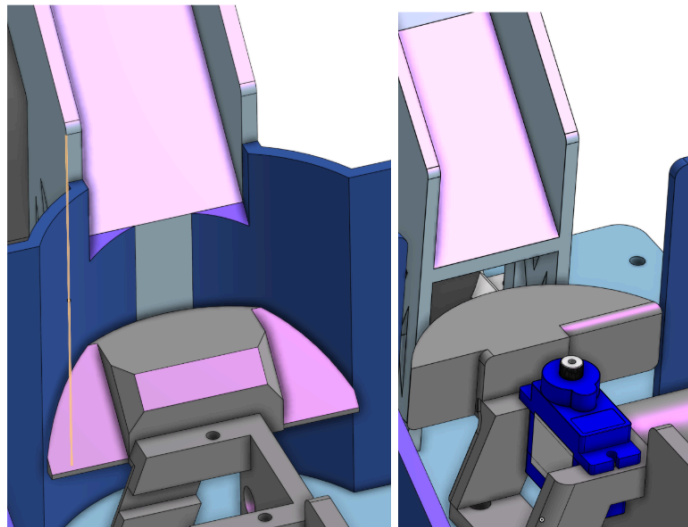


Figure 17: Implementation of Landing Platform Slope

2.2.7 Sorting System Software Implementation

The implementation aimed to determine threshold values for the TCS34725 colour sensor to reliably classify colours (red, green, blue, yellow, orange, purple, pink, black, and white) for a sorting mechanism, while maintaining robustness against lighting variation and sensor noise.

Raw sensor data consists of Red (R), Green (G), Blue (B), and Clear (C) channels, where C represents brightness. To improve stability, 10 samples are averaged using $r = rSum / cNumSamples$, $g = gSum / cNumSamples$, $b = bSum / cNumSamples$, and $c = cSum / cNumSamples$. This reduces noise, smooths lighting fluctuations, and improves repeatability.

To remove brightness dependence, RGB values are normalized by computing:

$$sum = r + g + b$$

$$r_{\square} = r / sum$$

$$g_{\square} = g / sum$$

$$b_{\square} = b / sum$$

This ensures colour detection is based on relative proportions rather than absolute intensity.

Experimental data was collected using normalized values and brightness (C), revealing clear patterns between colours. For example, orange showed $r_{\square} \approx 0.58$, $g_{\square} \approx 0.22$, $b_{\square} \approx 0.20$; pink showed $r_{\square} \approx 0.54$, $g_{\square} \approx 0.20$, $b_{\square} \approx 0.25$; green had $g_{\square} \approx 0.50$; white had balanced RGB with high brightness (~ 600); and black had very low brightness with balanced RGB.

Thresholds were derived from these patterns. Brightness based classification was applied first: black was defined as $C < 40$, and white as $C > 300$ with balanced RGB. Primary colours were detected using dominant channels, with green identified by $g_{\square} > 0.45$ and blue by $b_{\square} > 0.45$ with $C > 60$ to avoid low light errors.

Mixed colours were classified using combined thresholds: orange ($r_{\square} > 0.52$, $g_{\square} > 0.18$, $b_{\square} < 0.25$), pink ($r_{\square} > 0.50$, $b_{\square} > 0.22$, $g_{\square} < 0.30$), yellow ($r_{\square} > 0.32$, $g_{\square} > 0.35$, $b_{\square} < 0.28$), and purple ($b_{\square} > 0.35$, $r_{\square} > 0.28$, $g_{\square} < 0.32$).

To prevent overlap, a priority order was used: black/white to strong colours (green, blue) to mixed colours (yellow, orange, purple, pink) to red (default). This ensures more specific conditions are evaluated first.

After calibration, the system achieved accurate detection of all colours. Misclassifications (e.g., orange as red, yellow as unknown) were eliminated, and the system became robust to lighting changes (via normalization) and noise (via averaging).

Limitations include reduced accuracy at very low brightness, the need for closer proximity for blue due to low reflectivity, and sensitivity between similar colours (e.g., pink vs red) under varying lighting.

Overall, the implementation combines empirical calibration, normalization, and iterative threshold refinement to achieve reliable classification and integration with the sorting system. Improvements could include using HSV colour space, automatic calibration, dynamic threshold storage, and confidence scoring.

2.2.8 Pick Up & Drive Software Implementation

For the navigation system proposed in the preliminary design phase, it was determined late in the detailed design phase that the approach was not viable. The design required the field to be surrounded by walls, which was later confirmed to violate mission rules after consultation with the TA. As a result, the navigation strategy was dropped. Due to limited time remaining in the project, a simplified hardcoded approach was implemented. This method used a timing loop within the code to track elapsed time and controlled motion. Based on predefined time intervals obtained through trial and error, the robot was commanded to move by sending PWM signals directly to the motors.

However, this approach introduced positional error, as movement was based solely on time rather than sensor feedback. Variations in motor speed, surface friction, and interaction with erasers led to accumulated error in position tracking. As a result, complex movement paths were not reliable, and the robot could not consistently return to its starting position, violating constraint C-7.

To mitigate this, the motion strategy was simplified to a linear forward and back path. This reduced accumulated error and improved repeatability, as it minimized variability in travel distance and system response.

A similar approach was applied to the pick-up wheel. During testing, it was observed that erasers could become lodged between the floor and the lifting mechanism depending on their orientation, causing the wheel to stall. To address this, a “jitter” motion was implemented in the control logic. The wheel was programmed to rotate in the lifting direction for a fixed duration, followed by a brief reverse rotation (0.5 s). This reverse motion was sufficient to dislodge stuck erasers, allowing them to be repositioned and picked up in the next cycle. This approach improved reliability of the pick-up system without requiring additional mechanical changes.

2.2.9 Electrical Systems

It was identified that a single battery would likely be insufficient to supply adequate current to all components, including 4× N20 motors, 1× servo motor, 1× color sensor, and the required control hardware. This assumption was based on prior experience in MSE 2202 labs, where the battery showed performance issues under higher electrical loads, even when fully charged.

At the same time, the system architecture required the use of two ESP32 microcontrollers. This was due to difficulties in merging the driving/navigation code with the sorting system code, which led to unreliable behavior when combined. To address this, the systems were separated, with each ESP32 responsible for a

specific subsystem. While this introduced additional weight and negatively impacted Objective O-3, it was a necessary trade off to ensure reliable system operation within the project timeline.

As a result of both the increased electrical load and system separation, a dual battery configuration was implemented. One battery powered 1× ESP32 and 2× N20 motors, while the second battery powered 1× ESP32, 2× N20 motors, the servo motor, and the color sensor. This distribution reduced the load on each battery and improved overall.

3 Conclusion

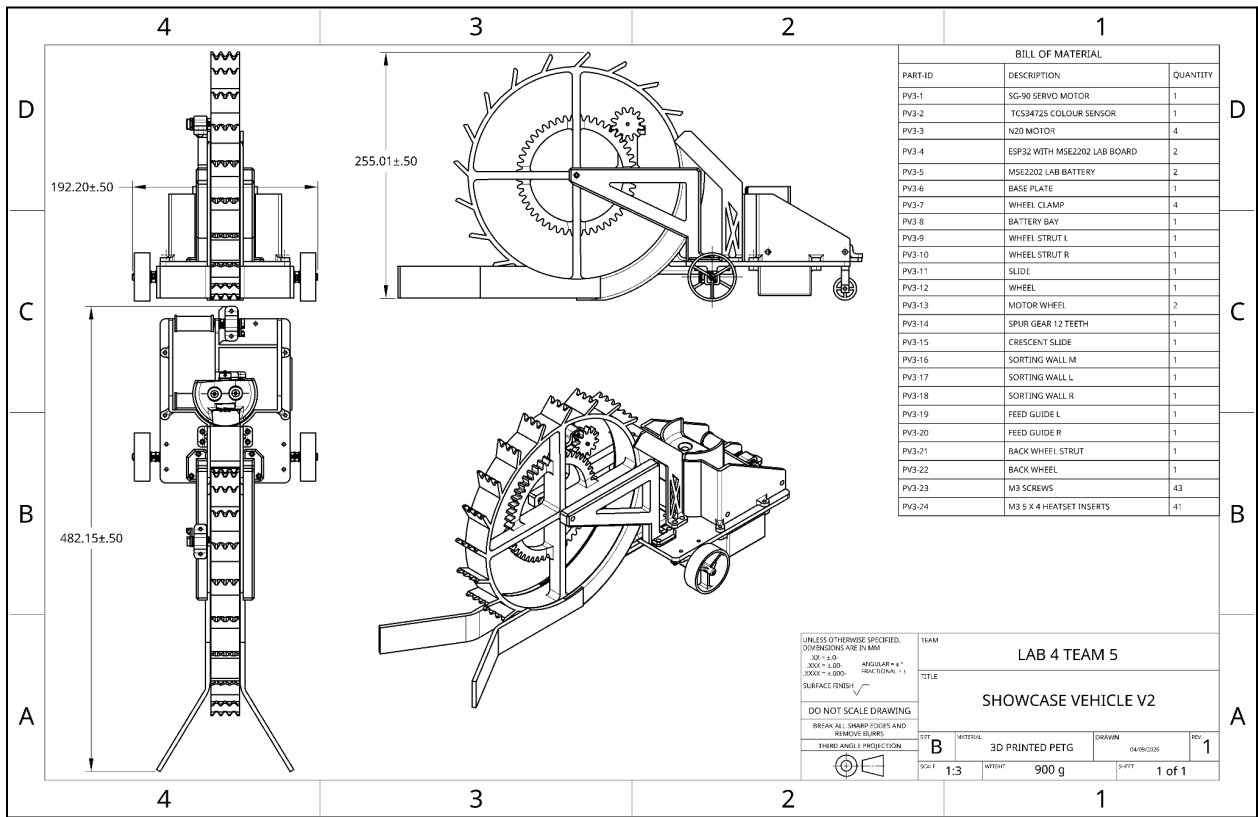


Figure 18: Technical Drawing of Final Design

The final design (Showcase Vehicle v2) had overall dimensions of 482 × 192 × 255 mm and a net weight of approximately 900 g, consisting of 17 custom 3D printed components. The project followed an

iterative design approach, where early design shortcomings were identified through testing and addressed in subsequent iterations.

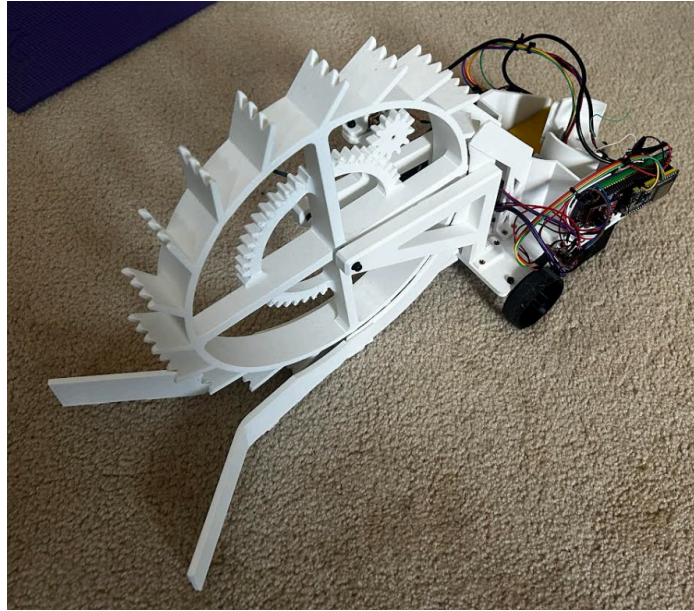


Figure 19: Implementation of Final Design

Overall performance at the showcase was below expectation, although most objectives and constraints were met. The primary limitation was in navigation. The vehicle was not able to operate fully autonomously, as it could not reliably return to the collection bin. Human assistance was required to reposition the bin for successful eraser deposition, resulting in a failure to meet constraint C-7 (autonomy) and Objective O-4 (navigation accuracy). This limitation was primarily due to the late stage shift to a time based control strategy, which introduced positional error and reduced reliability.

Despite this, the mechanical systems performed well. The pick-up system operated consistently, and after multiple design iterations, was able to reliably collect erasers without jamming. The sorting system was a major success, achieving 100% sorting accuracy during the showcase. This strongly satisfied Objective O-2, where sorting accuracy was prioritized over total collection quantity due to scoring penalties. Significant improvements were also made in manufacturability, serviceability, and assembly. The

transition to a modular design using M3 fasteners and heat-set inserts improved Objectives O-7 and O-8 by allowing components to be easily assembled, disassembled, and replaced without damage. Standardization across subsystems (Objective O-9) reduced integration issues and improved collaboration between team members. Objective O-6 (cost) was also met effectively. The final design used approximately 210 g of PETG filament (~\$3.78), along with 41 heat-set inserts (~\$4.47) and 43 M3 screws (~\$0.73), resulting in a total external cost of approximately \$8.98. No additional electronic components beyond those provided in the MSE 2202 lab kit were required. This kept the system well within budget while maintaining functionality.

Overall, the project demonstrated a strong application of iterative engineering design. While full autonomy was not achieved, the final system met most functional requirements and provided a reliable and well integrated solution for eraser collection and sorting within the given constraints.

4.1.0 Mechanical Improvements

The pick-up mechanism should be validated early using real test conditions. The failure of the stick based system was a major design oversight, as it relied on incorrect assumptions about material interaction. The eraser coating prevented reliable adhesion, which was only discovered after significant design effort.

Future designs should prioritize early validation of contact based mechanisms and favor more passive, geometry based solutions that are less sensitive to material properties.

4.2 Electrical & Software Improvements

The navigation system was the primary limitation of the final design. Future work should focus on implementing a feedback based navigation system rather than relying on time based control. Even a

simple closed loop system using encoders, line tracking, or boundary sensing would significantly improve positional accuracy and allow the system to meet autonomy requirements.

From a system integration perspective, the need to split control across two ESP32 units indicates that software architecture should be considered earlier. A more modular code structure or defined communication protocol between subsystems could allow for integration on a single microcontroller, reducing system complexity and weight.

4.3 Project Structure Improvements

One major limitation of the project was the lack of score based design evaluation during development. While scoring criteria were analyzed early on, they were not actively used to guide design decisions. Incorporating score prediction throughout the design process would allow for more informed trade offs, particularly between weight, speed, and sorting performance.

Additionally, insufficient time was allocated after manufacturing for testing and iteration. This resulted in rushed design changes and limited optimization. Future project timelines should allocate a dedicated buffer for post-manufacturing testing and refinement, allowing for more controlled iterations and better informed design decisions.